

non-homogenous linear systems

We will be dealing with undetermined coefficients and variation of parameters

our equation will have a complementary function and a particular function:

$$\left. \begin{array}{l} \text{complementary} \\ \text{Particular} \end{array} \right\} \begin{array}{l} x_c \\ x_p \end{array} \quad x = x_c + x_p$$

$$\frac{dx}{dt} = 6x + y + 6t$$

$$\frac{dy}{dt} = 4x + 3y - 10t + 4$$

• when you see a "t" in the equation you are dealing with a non-homogenous system

$$x' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} x + \begin{pmatrix} 6 \\ -10 \end{pmatrix} t + \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

• you will need to use undetermined coefficients to find a x_p for this

$$\det(A - \lambda I) = \begin{vmatrix} 6 - \lambda & 1 \\ 4 & 3 - \lambda \end{vmatrix}$$

$$(6 - \lambda)(3 - \lambda) - (1)(4)$$

$$\lambda^2 - 9\lambda + 14 = 0$$

$$(\lambda - 7)(\lambda - 2) = 0$$

$$\lambda_1 = 7, \lambda_2 = 2$$

$$@ \lambda = 7$$

$$\left[\begin{array}{cc|c} -1 & 1 & 0 \\ 4 & -4 & 0 \end{array} \right]$$

$$\xrightarrow{4R_1 + R_2 \rightarrow R_2}$$

$$\left[\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] \xrightarrow{-R_1}$$

$$\begin{array}{cc|c} x_1 & x_2 & \\ \hline 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}$$

$$\begin{aligned} x_1 - x_2 &= 0 \\ x_1 &= x_2 \\ (1) &= (1) \end{aligned}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{7t}$$

@ $\lambda = 2$

$$\begin{array}{cc|c} 4 & 1 & 0 \\ 4 & 1 & 0 \end{array}$$

$$\xrightarrow{R_1 - R_2}$$

$$\begin{array}{cc|c} 4 & 1 & 0 \\ 0 & 0 & 0 \end{array}$$

$$\xrightarrow{\frac{1}{4} R_1}$$

$$\begin{array}{cc|c} x_1 & x_2 & \\ \hline 1 & 1/4 & 0 \\ 0 & 0 & 0 \end{array}$$

$$\begin{aligned} x_1 - 1/4 x_2 &= 0 \\ x_1 &= \frac{1}{4} x_2 \\ (1) &= \frac{1}{4} (4) \end{aligned}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{2t}$$

$$x_c = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{7t} + C_2 \begin{bmatrix} -1 \\ 4 \end{bmatrix} e^{2t}$$

• we got x_c
now we need
to find x_p

Find x_p :

$$x_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} t + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

• Plug this into "x"

• we used undetermined coefficients to get x_p

$$x' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} x + \begin{pmatrix} 6 \\ -10 \end{pmatrix} t + \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} \left[\begin{pmatrix} a_1 \\ b_1 \end{pmatrix} t + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} \right] + \begin{pmatrix} 6 \\ -10 \end{pmatrix} t + \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = \begin{bmatrix} (6a_1 + b_1 + 6)t + (6a_2 + b_2 + 0) \\ (4a_1 + 3b_1 - 10)t + (4a_2 + 3b_2 + 4) \end{bmatrix}$$

these are all lined to "t" the ones with a "t" are like terms
so we can combine and factor

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{bmatrix} (6a_1 + b_1 + 6)t + 6a_2 + b_2 - a_1 \\ (4a_1 + 3b_1 - 10)t + 4a_2 + 3b_2 + 4 - b_1 \end{bmatrix}$$

we subtracted these to the other side

terms associated with "t"

$$(6a_1 + b_1 + 6) = 0$$

$$(4a_1 + 3b_1 - 10) = 0$$

$$a_1 = -2, \quad b_1 = 6$$

terms associated with "c"

$$6a_2 + b_2 - a_1 = 0$$

$$4a_2 + 3b_2 + 4 - b_1 = 0$$

$$a_2 = -417$$

$$b_2 = 1017$$

Plug the answer into the x_p equation:

$$x_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} t + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

$$x_p = \begin{pmatrix} -2 \\ 6 \end{pmatrix} t + \begin{pmatrix} -417 \\ 1017 \end{pmatrix}$$

write the general equation: $x = x_c + x_p$

$$x = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{7t} + c_2 \begin{bmatrix} -1 \\ 4 \end{bmatrix} e^{2t} + \begin{pmatrix} -2 \\ 6 \end{pmatrix} t + \begin{pmatrix} -417 \\ 1017 \end{pmatrix}$$

$$x' = \begin{pmatrix} 4 & 1/3 \\ 9 & 6 \end{pmatrix} x + \begin{pmatrix} -3 \\ 4 \end{pmatrix} e^t$$

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 1/3 \\ 9 & 6-\lambda \end{vmatrix}$$

$$(4-\lambda)(6-\lambda) - (9)(1/3)$$

$$\lambda^2 - 10\lambda + 21$$

$$(\lambda-3)(\lambda-7)$$

$$\lambda_1 = 3 \quad \lambda_2 = 7$$

$$@ \lambda = 3$$

$$\begin{vmatrix} 1 & 1/3 \\ 9 & 3 \end{vmatrix}$$

$$\xrightarrow{9R_1 - R_2}$$

$$\begin{vmatrix} x_1 & x_2 \\ 1 & 1/3 \\ 0 & 0 \end{vmatrix}$$

$$x_1 + 1/3 x_2$$

$$x_1 = -1/3 x_2$$

$$(-1) = -1/3 (3)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} e^{3t}$$

$$@ \lambda = 7$$

$$\begin{vmatrix} -3 & 1/3 \\ 9 & 3 \end{vmatrix}$$

$$\xrightarrow{R_1 + \frac{1}{3} R_2}$$

$$\begin{vmatrix} -3 & 1/3 \\ 0 & 0 \end{vmatrix} \xrightarrow{-\frac{1}{3} R_1}$$

$$\begin{vmatrix} x_1 & x_2 \\ 1 & -1/9 \\ 0 & 0 \end{vmatrix}$$

$$x_1 - \frac{1}{9} x_2 = 0$$

$$x_1 = \frac{1}{9} x_2$$

$$(1) = \frac{1}{9} (9)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \end{bmatrix} e^{7t}$$

$$x_c = c_1 \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 9 \end{bmatrix} e^{7t}$$

• we have found x_c
we now need to find x_p

Find x_p :

$$x_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} e^t$$

Plug the x_p into the equation:

$$x' = \begin{pmatrix} 4 & 1/3 \\ 9 & 6 \end{pmatrix} x + \begin{pmatrix} -3 \\ 4 \end{pmatrix} e^t$$

$$\begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = \begin{pmatrix} 4 & 1/3 \\ 9 & 6 \end{pmatrix} \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} e^t + \begin{pmatrix} -3 \\ 4 \end{pmatrix} e^t$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{bmatrix} (4a_1 + \frac{1}{3}b_1 - 3)e^t \\ (9a_1 + 6b_1 + 4)e^t \end{bmatrix} - \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

$$\begin{cases} 3a_1 + \frac{1}{3}b_1 = 3 \\ 9a_1 + 5b_1 = -4 \end{cases}$$

$$b_1 = \frac{-13}{4}, \quad a_1 = \frac{49}{36}$$

Plug this into the x_p equation:

$$x_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} e^t$$

$$x_p = \begin{pmatrix} 49/36 \\ -13/4 \end{pmatrix} e^t$$

$$x = c_1 \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 9 \end{bmatrix} e^{7t} + \begin{pmatrix} 49/36 \\ -13/4 \end{pmatrix} e^t$$

We will now begin doing the Variation of Parameters part:

$$x' = \begin{pmatrix} -6 & 2 \\ -3 & 1 \end{pmatrix} x + \begin{pmatrix} e^t \\ t \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -6 - \lambda & 2 \\ -3 & 1 - \lambda \end{vmatrix} \quad \begin{aligned} &(-6 - \lambda)(1 - \lambda) - (-3)(2) \\ &\lambda_1 = 0, \lambda_2 = -5 \end{aligned}$$

@ $\lambda_1 = 0$:

$$\left[\begin{array}{cc|c} -6 & 2 & 0 \\ -3 & 1 & 0 \end{array} \right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{cc|c} -6 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right] \xrightarrow{-1/6 R_1}$$

$$\begin{array}{cc} x_1 & x_2 \\ \left[\begin{array}{cc|c} 1 & -1/3 & 0 \\ 0 & 0 & 0 \end{array} \right] \end{array}$$

$$\begin{aligned} x_1 - 1/3 x_2 &= 0 \\ x_1 &= 1/3 x_2 \\ (1) &= 1/3 (3) \end{aligned}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

@ $\lambda = -5$

$$\left[\begin{array}{cc|c} -1 & 2 & 0 \\ -3 & 6 & 0 \end{array} \right] \xrightarrow{3R_1 - R_2} \left[\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right] \xrightarrow{-R_1}$$

$$\begin{array}{cc} x_1 & x_2 \\ \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array} \right] \end{array}$$

$$\begin{aligned} x_1 - 2x_2 &= 0 \\ x_1 &= 2x_2 \\ (2) &= 2(1) \end{aligned}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t}$$

$$x_c = C_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + C_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-st}$$

• we have found the x_c now we need to find x_p

For the next section we need to know two equations:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$x_p = \Phi(t) \int \Phi^{-1}(t) F(t)$$

you get the $\Phi(t)$ equation from x_c :

$$\Phi(t) = \begin{bmatrix} 1 & 2e^{-st} \\ 3 & e^{-st} \end{bmatrix}$$

now we need to find the inverse of $\Phi(t)$:

$$\Phi^{-1}(t) = \frac{1}{e^{-st} - 6e^{-st}} \begin{bmatrix} e^{-st} & -2e^{-st} \\ -3 & 1 \end{bmatrix}$$

$$\Phi^{-1}(t) = \frac{1}{-5e^{-st}} \begin{bmatrix} e^{-st} & -2e^{-st} \\ -3 & 1 \end{bmatrix}$$

• now distribute the $\frac{1}{-5e^{-st}}$ into the matrix

$$\Phi^{-1}(t) = \begin{bmatrix} -1/5 & 2/5 \\ \frac{3}{5}e^{st} & -\frac{1}{5}e^{st} \end{bmatrix}$$

now we will find $\Phi^{-1}(t) F(t)$, $F(t)$ is $\begin{bmatrix} e^t \\ t \end{bmatrix}$ found in the original equation

$$\Phi^{-1}(t) F(t) = \begin{bmatrix} e^t \\ t \end{bmatrix} \begin{bmatrix} -1/5 & 2/5 \\ \frac{3}{5}e^{5t} & -\frac{1}{5}e^{5t} \end{bmatrix}$$

$$\Phi^{-1}(t) F(t) = \begin{bmatrix} -\frac{e^t}{5} + \frac{2}{5}t \\ \frac{3}{5}e^{6t} - \frac{1}{5}te^{5t} \end{bmatrix}$$

We are following the equation $\Phi(t) \int \Phi^{-1}(t) F(t)$ so that means you need to take an integral of $\Phi^{-1}(t) F(t)$

$$\int \begin{bmatrix} -\frac{e^t}{5} + \frac{2}{5}t \\ \frac{3}{5}e^{6t} - \frac{1}{5}te^{5t} \end{bmatrix}$$

integration by parts

uv - $\int v du$

I'll just write the answer

$$\Rightarrow \begin{bmatrix} -\frac{e^t}{5} + \frac{1}{5}t^2 \\ \frac{1}{10}e^{6t} - \frac{t}{25}e^{5t} + \frac{1}{125}e^{5t} \end{bmatrix}$$

now all we need to do is this:

$$\Phi(t) \int \Phi^{-1}(t) F(t)$$

$$\begin{bmatrix} 1 & 2e^{-5t} \\ 3 & e^{-5t} \end{bmatrix} \begin{bmatrix} -\frac{e^t}{5} + \frac{1}{5}t^2 \\ \frac{1}{10}e^{6t} - \frac{t}{25}e^{5t} + \frac{1}{125}e^{5t} \end{bmatrix}$$

$$\begin{bmatrix} \left(-\frac{e^t}{5} + \frac{1}{5}t^2\right) \\ \left(\frac{1}{10}e^{6t} - \frac{t}{25}e^{5t} + \frac{1}{125}e^{5t}\right) \end{bmatrix} \begin{bmatrix} 1 & 2e^{-5t} \\ 3 & e^{-5t} \end{bmatrix}$$

$$\begin{bmatrix} \left(-\frac{e^t}{5} + \frac{1}{5}t^2\right) + \left(\frac{1}{5}e^t - \frac{2t}{25} + \frac{2}{125}\right) \\ \left(-\frac{3}{5}e^t + \frac{3}{5}t^2\right) + \left(\frac{1}{10}e^t - \frac{1}{25}t + \frac{1}{125}\right) \end{bmatrix}$$

you can simplify this:

$$x_p = \begin{bmatrix} \frac{1}{5}t^2 - \frac{2}{25}t + \frac{2}{125} \\ -\frac{1}{2}e^t + \frac{3}{5}t^2 - \frac{1}{125}t + \frac{1}{125} \end{bmatrix}$$

now we combine x_c and x_p into one

$$x = C_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + C_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t} + \begin{bmatrix} \frac{1}{5}t^2 - \frac{2}{25}t + \frac{2}{125} \\ -\frac{1}{2}e^t + \frac{3}{5}t^2 - \frac{1}{125}t + \frac{1}{125} \end{bmatrix}$$